السلام عليكم ورحمة الله وبركاته.. وبعد،

بين يديك ملخص شامل ومنظم لمادة CPCS-241 _ Chapter 3، تم إعداده بعناية ليكون عوناً لك في فهم المادة .وقبل البدء، أضع بين يديك ميثاقاً للأمانة العلمية والتعاون:

⚠️ ميثاق الأمانة والمسؤولية:

هذا العمل هو جهد طالب، أضعه بين يديك بنية النفع والفائدة؛ لذا:

- لا أسامح ولا أحلل استخدامه في الغش أو أي عمل ينافي الأخلاق الإسلامية.
- يُمنع منعاً باتاً تداوله كمصدر تجاري للباعة أو استخدامه في أدوات الذكاء الاصطناعي لغرض التحايل او الغش.
- الأمانة العلمية تقتضي منك استخدامه كمرجع للفهم والمراجعة، لا كبديل عن الضمير والاجتهاد الشخصي.

🌐 خدمات إضافية عبر الموقع:

يمكنك تعزيز مذاكرتك من خلال الوصول إلى أسئلة تفاعلية تجريبية للمنهج عبر الرابط: 🔗 kau-one.web.app/quiz

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📣 التواصل معي:

الكمال لله وحده، لذا يسعدني استقبال ملاحظاتكم، تعديلاتكم، أو إضافاتكم عبر:

- الرقم المباشر: 0530717788
- الموقع الشامل: kau-one.web.app

توجيه إيماني:

قبل أن تبدأ، استعذ بالله من الشيطان الرجيم، وجدد نيتك بأن يكون هذا الوقت في سبيل طلب العلم النافع الذي يرفع قدرك في الدنيا والآخرة. قال رسول الله ﷺ: "مَنْ سَلَكَ طَرِيقًا يَلْتَمِسُ فِيهِ عِلْمًا سَهَّلَ اللَّهُ لَهُ بِهِ طَرِيقًا إِلَى الْجَنَّةِ" (رواه مسلم).

إعداد الطالب: إلياس حسان بابور الدفعة: 24 — تم اخذ الماده و عمل الملخص على السلايدات او تم اخذ الماده 2026

The Design Process (The Strategy)

Before you build a house, you draw the blueprint. Database design has two main sides:

1. **Database Design:** Designing the actual data structure (The "Conceptual Schema"). This is our focus.
2. **Applications Design:** Designing the programs (the buttons and screens) that use the data. This is more about Software Engineering.

The "Mini-World" (The Requirements)

like the COMPANY example and told to **draw it**. To do this, you must learn to "get" Entities and Attributes from the text.

The COMPANY List:

- **DEPARTMENT:** Has a name, number, and locations.
- **PROJECT:** Has a name, number, and location.
- **EMPLOYEE:** Has SSN, address, salary, sex, birthdate.
- **DEPENDENT:** Has name, sex, birthdate, relationship.

The Core Concepts (The Tools)

1. Entities

- **Definition:** An object or "thing" in the mini world (e.g., a specific employee).
- **Drawing:** Represented by a **Rectangle**.

2. Attributes

- **Definition:** The properties that describe the entity (e.g., the Name of the employee).
- **Drawing:** Represented by an **Oval** connected to the Entity.
- **Value Sets:** Every attribute has a "Domain" (like a data type: Integer, String, Date).

Attribute Flavors & Keys

1. The Three Main Types

- **Simple (Atomic):** The basic level. One value only (e.g., Sex: 'M').
    - **Drawing:** A single Oval.
- **Composite:** An attribute that can be broken into smaller parts (e.g., Name breaks into First_Name, Last_Name).
    - **Drawing:** An Oval that sprouts "branches" of other Ovals.
- **Multi-valued:** An entity can have *many* values for this (e.g., a Car having many Colors, or a person having two Phone Numbers).
    - **Drawing:** A **Double Oval**.

2. The Key Attribute

This is the "Unique ID" of your entity. No two entities can have the same value for a Key (like your SSN).

- **Fact:** An entity can have **more than one key**.
- **Fact:** A key can be **composite** (e.g., a Car's License Plate is made of 'Number' and 'State').
- **Drawing:** The text inside the Oval is **<u>Underlined</u>**.

Note: this is different from the relational schema where only one “primary key is underlined

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The Collection and the Constraints

Now we look at the "Entity Set" and the "Value Set." Think of these as the **Container** and the **Rules**.

1. Entity Set (The Container)

- **What it is:** While the **Entity Type** is the blueprint (e.g., "STUDENT"), the **Entity Set** is the actual group of all students currently in the system.
- **Confusing Point:** Usually, we use the same name for both. If I say "The EMPLOYEE table," I’m usually talking about the Entity Set (the current state of all employees).

2. Value Sets / Domains (The Rules)

- **Definition:** This is essentially the **Data Type**.
- **The "P(V)":** $A : E \rightarrow P(V)$. This is as saying an attribute (A) takes an entity (E) and gives you a value (or a set of values, like in multi-valued attributes) from the allowed "Value Set" (V).
- **Examples:** Age must be an Integer. Name must be a String.

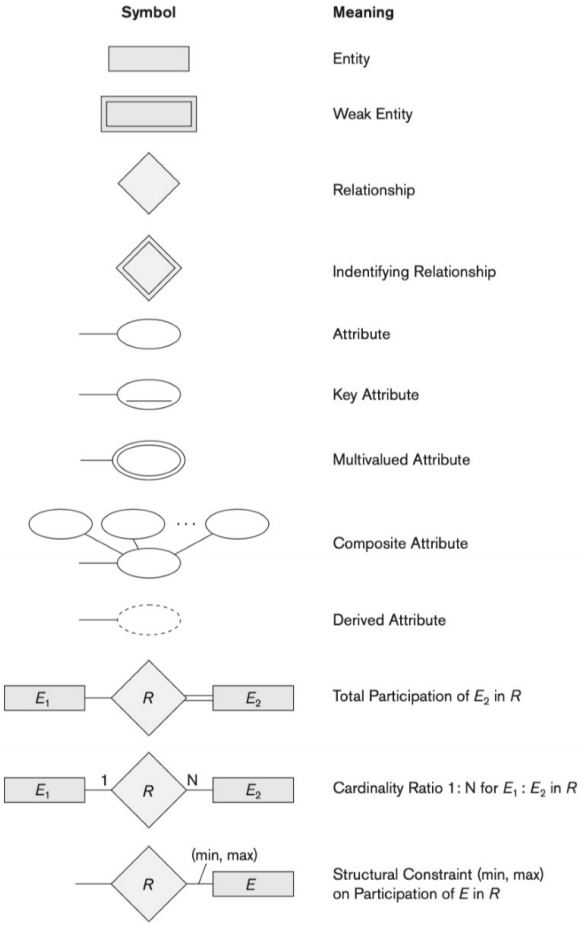

Figure 3.14
Summary of the notation for ER diagrams.

Relationships & Connectivity

1. What is a Relationship?

- **Relationship Type:** The "blueprint" of how entities interact (e.g., **WORKS_FOR**).
- **Relationship Instance:** A specific connection (e.g., "elyas works for the CIA").
- **Degree:** The number of entity types involved.
    - **Binary:** 2 entities (most common).
    - **Ternary:** 3 entities.
    - and of degree n are called n-ary
- In general, an n-ary relationship is not equivalent to n binary relationships

2. How to Draw It (The Diamond Strike)

- **The Shape:** A **Diamond** ♢.
- **The Connection:** Draw a straight line from the Entity (Rectangle) to the Relationship (Diamond).

- **The Reading Rule:** Names should be readable from **left-to-right** or **top-to-bottom**. No arrows needed yet!

note : when a relation is linked to the same entity then it's called : Recursive

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Type vs. Set

Think of it like a **Facebook Relationship Status**:

- **Relationship Type (The Blueprint):** This is the "logic" or the "box." It says: *"A User can be 'In a Relationship' with another User."* This is what you draw in your diagram (the Diamond).
- **Relationship Set (The Data):** This is the list of everyone actually dating right now.
    - Instance 1: (Ali, Sarah)
    - Instance 2: (John, Mary)

Constraints & Recursion

Constraints (the numbers and lines).

1. Cardinality Ratio (The "How Many" Rule)

This tells us the maximum number of relationship instances an entity can participate in.

- **1:1 (One-to-One):** One Manager manages one Department.
- **1:N (One-to-Many):** One Department has many Employees.
- **M:N (Many-to-Many):** Many Employees work on many Projects.

2. Participation Constraint (The "Must You?" Rule)

- **Total Participation (Mandatory):** Every entity *must* be in the relationship.
    - **Drawing:** Represented by a **Double Line** connecting the Entity to the Relationship.
- **Partial Participation (Optional):** Some entities might not participate.
    - **Drawing:** Represented by a **Single Line**.

3. Recursive Relationships (The Mirror)

When an entity relates to itself (like an Employee supervising another Employee).

- **Key Detail:** You **must** write the **Role Names** on the lines (e.g., "Supervisor" and "Subordinate") so we know who is who!

The Two Pillars of Constraints

1. **Cardinality (Max Participation):** Tells you the *maximum* connections. You draw **1, N, or M** on the line.
2. **Participation (Min Participation):** Tells you if it is *required* or *optional*.
    - **Total (Required):** Draw a **Double Line**.
    - **Partial (Optional):** Draw a **Single Line**.

(Weak Entities & Ternary)

1. Weak Entities (The Dependent)

- **What it is:** An entity with no unique key of its own (e.g., a "Dependent" named 'Sarah'). There might be many Sarahs in the company, so we only know *which* Sarah by looking at the Employee she belongs to.
- **Partial Key:** The attribute that distinguishes weak entities *under the same owner* (e.g., Name).
- **Drawing the Weak Entity:** check the table photo upove

2. Relationship Attributes

- Yes, diamonds can have ovals! If an attribute only exists when two entities meet (like `Hours` an employee works on a specific project), you attach that oval to the **Diamond**.

3. (Min, Max) Notation

- This is an alternative way to show constraints. Instead of 1:N and double lines, you write `(min, max)` next to the entity.
- **Example:** `(1, 1)` means exactly one. `(0, n)` means optional, many.

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info that might come as true false maybe

Appendix A illustrates some of the alternative notations that have been used UML class diagrams is representative of another way of displaying ER concepts that is used in several commercial design tool

In some cases, a ternary relationship can be represented as a weak entity if the data model allows a weak entity type to have multiple identifying relationships (and hence multiple owner entity types)

Redundancy: If you can get the same information from a Ternary relationship that you get from a Binary one, the Binary one is **redundant** (useless) and should be removed.

(this is a repeated info just for declaration)

the Weak Entity

Imagine you have an **EMPLOYEE** named elyas. elyas has a **DEPENDENT** (a son) named "Ali."

- **The Problem:** Is "Ali" a unique key? No. There are millions of Alis in the world.
- **The Solution:** Ali only becomes "unique" in our database when you pair him with elyas.
- **The Partial Key:** We call Ali's name a **Partial Key** (also called a *Discriminator*). It distinguishes between *elyas's* children (Ali vs. Sara), but it's not enough to find him in the whole company without knowing he belongs to elyas.

ملاحظة ختامية

الحمد لله الذي بنعمته تتم الصالحات، والصلاة والسلام على معلم البشرية الخير، نبينا محمد وعلى آله وصحبه أجمعين.

تم بفضل الله وتوفيقه الانتهاء من نهائيه الملخص

رسالة ختامية: كما ذكرتُ في المقدمة، فإن هذا الجهد أمانة بين يديك؛ فلا تجعل استخدامه يخرج عن إطار طلب العلم ، وابتعد عن كل ما يخدش ضمير الطالب المسلم من غش أو استغلال غير أخلاقي.

تذكير أخير: ما كان في هذا الملخص من صواب فمن الله وحده، وما كان فيه من خطأ أو نسيان فمني ومن الشيطان. وكما هو موضح في بيانات التواصل، بابي مفتوح لكل تصحيح أو إضافة تثري هذا العمل وتخدم الدفعات القادمة.

دعاء: اللهم انفعنا بما علمتنا، وعلمنا ما ينفعنا، وزدنا علماً. اللهم اجعل ما قرأناه حجة لنا لا علينا، ووفقنا جميعاً في اختباراتنا وفي حياتنا العملية لخدمة دينه ووطننا.

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تم دراسه الماده 2026

تم بحمد الله.